

FORM 2
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COMPLETE SPECIFICATION
(See section 10 and M.13)

LUNCH BOX WITH INTEGRATED SAUCE DISPENSER

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The following specification particularly describes the invention and the manner in which it is to be performed

Lunch Box with Integrated Sauce Dispenser

Title: Lunch Box with Integrated Sauce Dispenser

Field of the Invention

The present invention relates to food storage containers, specifically a lunch box designed to store and dispense sauces conveniently.

Background of the Invention

Traditional lunch boxes often lack a convenient way to store and dispense sauces without the risk of leakage or contamination. Existing solutions typically involve separate containers that can be bulky and prone to spillage. The need for a more integrated and user-friendly solution is evident, particularly for individuals' on-the-go.

US20020084206 - Lunch box. The lunch box comprising a plurality of stackable, interchangeable food tray assemblies that have compartments in them for the storage of foodstuffs and other items. These stackable, interchangeable food tray assemblies are housed in an enclosure with a lid to protect the contents and a handle to provide ease of transport by the user. The lunch box can be made in a variety of sizes to accommodate the needs of the user and to fit within various size refrigerators. The lunch box can include locking features to hinder unauthorized access to the contents.

WO2020260961 - A MODULAR LUNCH BOX. A modular lunch box is disclosed, comprising a base to receive containers to store food articles and having one or both of heating and cooling devices; and a top lid removably received on base to cover and individually seal the containers. A control module is detachably received through a first aperture in top lid such that control module rests against a step portion of containers for operative coupling with heating and cooling devices to control their temperature. Without control module, lunch box is usable as a conventional lunch box. Control module includes a wireless communication module to receive user inputs to configure the lunch box for heating and/or cooling of food article at desired times, to desired temperatures. A battery pack positioned over top lid, gets coupled to control module to meet power requirement.

This invention is to integrate the sauce dispenser within the lunch box allowing users to store carry, and dispense sauces. The overall design provides a portable, versatile solution for meal storage and sauce dispensing, integrating both functionalities in a compact form factor. This eliminates the need for carrying separate sauce containers, reducing the number of items required for a meal and enhancing convenience for the user.

Summary of the Invention

The invention provides a lunch box (100) that integrates a sauce dispenser (101) into its design, allowing users to store, carry, and dispense sauces easily with separate container integrated within the box. The lunch box (100) features a compartment for solid food items and a built-in, leak-proof sauce dispenser that can be accessed easily when needed.

Detailed Description of the Invention

Structure of the Lunch Box (100): The lunch box (100) comprises a main body, a lid, and a sauce dispenser (101) module. The main body is divided into two compartments: one for solid food items and another for sauce storage. The sauce dispenser (101) module is attached to the main body, seamlessly integrating into the lunch box's design.

Sauce Dispenser Module (101): The dispenser is made of flexible, food-safe material that allows for easy squeezing to dispense sauces with a button (102) on top of the lunch box (100). A one-way valve at the dispenser's nozzle (101) prevents leakage while allowing for controlled dispensing of sauces. The module includes a removable container for easy cleaning and refilling. The sauce compartment which is integrated and insulated inside the top lid of the lunch box (100) to maintain the sauce's temperature and freshness.

Portability Features: The design is compact and lightweight, making it suitable for various activities such as school, work, or picnics. Users can customize the lunch box (100) with different color, designs, and sizes to suit individual preferences. Additional features may include microwave-safe materials, dishwasher-safe components, and eco-friendly materials.

Brief Description of the Drawings

The disclosed invention pertains to a lunch box (100) with an integrated sauce dispenser (101, 102) that allows for convenient storage, transportation, and dispensing of sauces. The lunch box (100) is composed of a rectangular, two-part structure, distinguished by a top cover with centrally positioned cylindrical knob that act as holder (103) and a bottom compartment to store food items.

The top cover is a rectangular lid with centrally positioned cylindrical knob that act as holder (103) and smooth, rounded edges, providing a secure seal over the main compartment. Positioned on the upper left corner of the lid, is a cylindrical dispensing knob (102) that serves as the activation mechanism for the integrated sauce dispenser (101). The knob (102) features a press down to dispense the sauce. This knob (102) could be connected to an internal pump mechanism that draws the sauce from a reservoir inside the lunch box. The front side of the compartment includes a vertically-aligned component that serves as a locking latch or clasp (104). This latch (104) secures the lid to the body, ensuring the contents are safely enclosed during transportation. The latch (104) is ergonomically shaped with rounded ends for ease of use, allowing for quick opening and closing of the lunch box. Internally, the top compartment includes a hidden reservoir directly connected to the dispensing knob (102) on the top cover, allowing the sauce to be dispensed (101) without opening the lunch box.

Integrated Sauce Dispenser (101): The sauce dispenser (101) is uniquely integrated into the lid of the lunch box (100), enabling users to dispense condiments directly onto their food without the need for separate containers. The dispenser operates through the cylindrical knob (102) on the top cover, which is connected to a container inside the lunch box, directing the sauce in a controlled manner. This design allows for one-handed operation, making it user-friendly and practical, especially for outdoor or on-the-go usage.

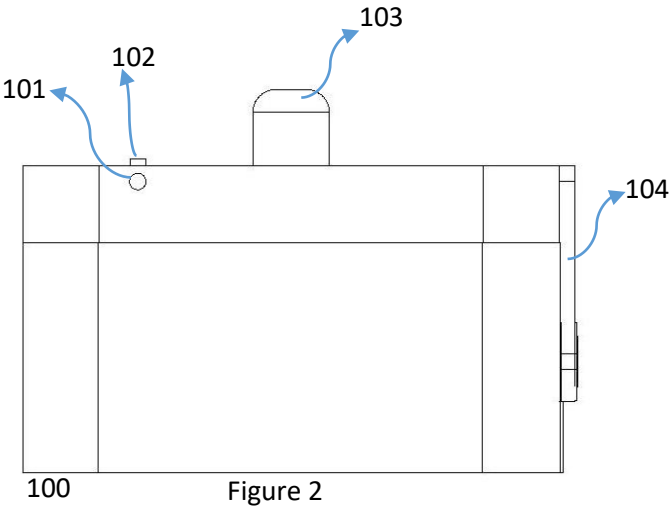
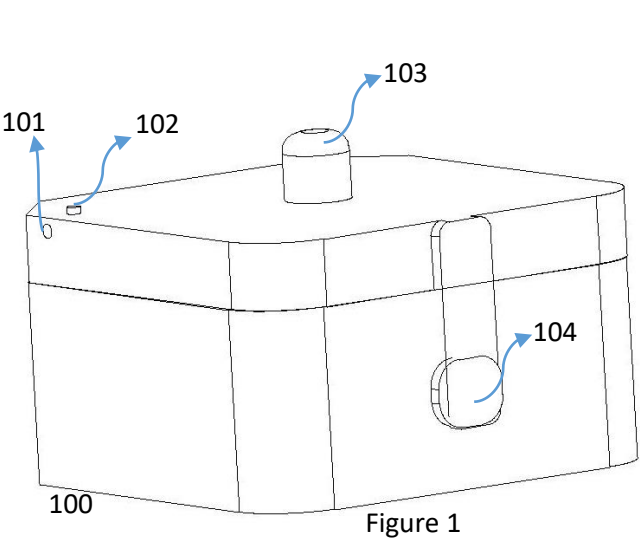
Additional Features: The lunch box's (100) design ensures that all components are leak-proof and spill-resistant, providing a clean and convenient solution for carrying meals and sauces together. The lunch box (100) is constructed from durable materials suitable for repeated use and easy cleaning, with smooth surfaces that prevent residue build-up.

We Claim

1. A lunch box (100) comprising a main body divided into two compartments, one for solid food items and another for storing sauces, with a built-in knob (102) operated sauce dispenser (101) module;
2. The knob (102) operated sauce dispenser (101) module of claim 1, wherein the dispenser (101) includes a flexible, food-safe material, a one-way valve, and a removable cap;
3. The lunch box (100) of claim 1, wherein A sauce dispensing (101) mechanism integrated into the top cover, the dispensing mechanism comprising of a central cylindrical knob (102) positioned on the exterior surface of the top cover, the knob (102) operable to control the flow of sauce;
4. The lunch box (100) of claim 1, wherein an internal sauce reservoir housed within the top cover, connected to the central cylindrical knob (102) through a nozzle or dispensing tube (101); the knob (102), when activated, dispenses sauce from the internal reservoir through a controlled outlet (101) positioned on the top cover, the central cylindrical knob (102) is configured to operate through a pressing motion, activating an internal pump mechanism to dispense (101) sauce in measured quantities;
5. The lunch box (100) of claim 1, further comprising a latch (104) mechanism positioned on the front wall of the base compartment, the latch (104) configured to securely lock the top cover in place and prevent accidental opening during transportation, the internal sauce reservoir is detachable from the top cover, allowing for easy filling, cleaning, and maintenance of the sauce storage area.
6. The lunch box (100) of claim 1, wherein the entire structure is constructed from food-grade materials that are dishwasher safe, heat-resistant, and leak-proof, making it suitable for repeated use and easy cleaning.

Date: 07/10/2024

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Abstract

The present invention relates to a lunch box (100) with an integrated sauce dispenser (101), designed to provide convenient storage, transportation, and dispensing of sauces directly onto food items without the need for separate containers. The lunch box comprises a two-part structure, including a top cover and a bottom compartment. The top cover is equipped with a centrally positioned cylindrical knob that serves as the holder (103) for the grip of the lunch box (100). The bottom compartment, designed for storing food items, includes a secure latch (104) to fasten the lid, ensuring a leak-proof seal during transportation. The internal sauce reservoir is connected to the dispensing knob (102), allowing users to dispense (101) sauces in a controlled manner without opening the lunch box. The device is ergonomic, compact, and constructed from durable materials suitable for repeated use, making it ideal for on-the-go meals and enhancing user convenience by combining food storage and sauce dispensing functionalities in a single, integrated solution.

Date: 07/10/2024

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